

Haas Inspection post – Calibration Data and Commissioning Mode

Haas Inspection Post Processor

The Haas post will need calibration macro numbers filled in before use. These values are entered in the Post properties fields when post processing from Fusion. Below is a brief explanation of the properties which should be set or confirmed.

There are 3 macro variables that define calibration data. Using Renishaw as an example, look for these 3 values, normally around macro variable #556.

- The calibrated radius will be just less than the actual radius (Renishaw default #556 and #557)
- The eccentricity (run-out in X) will be close to 0 (default #558)
- The eccentricity (run-out in Y) will also be close to 0 (default #559)

If the controller does not have values in macro variable #556, check for values as described above: two values just below the calibrated probe radius and two values very close to 0. In the Haas Next Generation controller macro variables #500 and #10500 are equivalent. If macro variables #556, #557, #558, and #559 are all 0 then you should look for the variable values matching the described pattern, or else contact the Renishaw team to know the exact variables.

Current Commands																					
Devices		Timers		Macro Vars		Active Codes		Tools		Plane		Calculator									
Macro Variables																					
(Local) 1 - 33				(Global) 10000 - 10199				(Global) 10200 - 10399													
Var	Value			Var	Value			Var	Value												
1				10000	0.000000			10200	0.000000												
2				10001	0.000000			10201	0.000000												
3				10002	0.000000			10202	0.000000												
4				10003	0.000000			10203	0.000000												
5				10004	0.000000			10204	0.000000												
6				10005	0.000000			10205	0.000000												
7				10006	0.000000			10206	0.000000												
8				10007	0.000000			10207	0.000000												
9				10008	0.000000			10208	0.000000												
10				10009	0.000000			10209	0.000000												
11				10010	0.000000			10210	0.000000												
12				10011	0.000000			10211	0.000000												
13				10012	0.000000			10212	0.000000												
14				10013	0.000000			10213	0.000000												
15				10014	0.000000			10214	0.000000												
16				10015	0.000000			10215	0.000000												
17				10016	0.000000			10216	0.000000												
18				10017	0.000000			10217	0.000000												
19				10018	0.000000			10218	0.000000												
20				10019	0.000000			10219	0.000000												

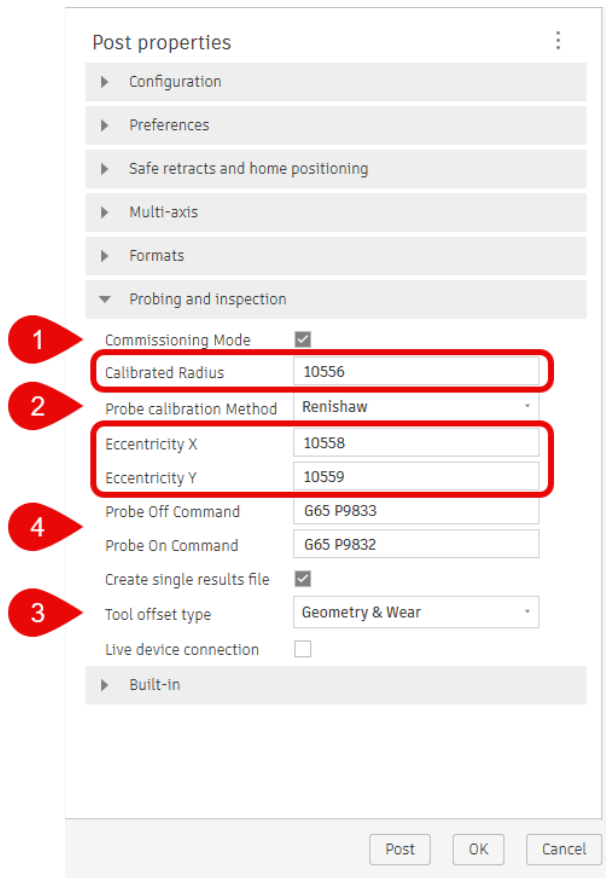
*Legacy 3 digit macros begin at 10000 Range. i.e. Macro 100 and 10100 are equivalent

A screenshot of these items highlighted in Fusion's post properties is shown in the Post Processor Settings section with some sample values. Once they are filled in with the correct macro variable values and commands, commissioning mode can be set to false, and then inspection programs can be posted.

To double check if you are looking at the proper variables, run the probe calibration process and see if the values are slightly changed.

Post Processor Settings

Here are some of the more common post-processor settings that you will need to pay attention to when post-processing your Inspection toolpath



The screenshot shows the 'Post properties' dialog box with the 'Probing and inspection' section expanded. Red callouts 1 through 4 point to the following settings:

- 1** Commissioning Mode: ☒
- 2** Calibrated Radius: 10556
- 3** Probe calibration Method: Renishaw
- 4** Eccentricity X: 10558
Eccentricity Y: 10559

Other visible settings include:

- Probe Off Command: G65 P9833
- Probe On Command: G65 P9832
- Create single results file: ☒
- Tool offset type: Geometry & Wear
- Live device connection: ☐

Buttons at the bottom: Post, OK, Cancel.

- 1. Commissioning Mode:** Enables M0 and messages at key points in the program. You can disable *Commissioning Mode* after a successful run.
- 2. Probe calibration Method:** The probe tip radius can be calibrated as a single value or two values (in X&Y direction). If Renishaw is selected as a calibration method, then the probe tip radius will be calculated as an average value of two numbers. *Calibrated Radius* #500 variable will be taken as the first value and the next variable #501 will be taken as the second.
- 3. Tool offset type:** Depends on your tool table type:
 - Memory A – single column to store geometry and wear. You should choose *Geometry only* (GEOMETRY ONLY)
 $\#8=\#[2000 + \#4111]$
 - Memory B or C – two columns or more. You should choose *Geometry and Wear* (GEOMETRY AND WEAR)
 $\#8=\#[2000 + \#4111] + \#[2200 + \#4111]$
- 4. Probe On/Off Command:** It can be an M-code or a subprogram call. Check that the probe flashes green after the probe on command has been executed. Renishaw Inspection+ cycles probe On and probe Off command will typically be G65 P9832 and G65 P9833. Check with the Machine documentation if needed.

Commissioning Mode

Inspection posts for Fusion defaults to Commissioning Mode. When Commissioning Mode is enabled, it will print messages on the controller screen at certain places throughout the inspection process for safety purpose.

The following checks are common while running in commissioning mode.

1. Ensure the probe is enabled.
 - a. The probe should move to the first point.
 - b. Verify the probe is enabled, and if so, press Cycle Start to continue.
2. Probe is about to contact part. Move should stop on contact.
 - a. Probe will position to the measuring point.
 - b. Make sure the probe is at the correct location and is about to make contact with the surface.
 - c. If it is positioned correctly press Cycle Start to continue.
 - d. Probe will touch the measuring surface and retract to a safe point.
3. Ensure the probe has been disabled.
 - a. The probe should move to a safe Z position.
 - b. Make sure the probe is disabled and press Cycle Start to continue.
4. The Results file can now be found in the same location as your NC code file or the folder specified in the *Results file location* property.
 - a. The Results file will be saved in the specified file folder.
 - b. Press Cycle Start to continue.
 - c. The Z-axis will now move to the machine's maximum Z location.

It is always recommended to enable Commissioning Mode the first time you run an inspection program. After the first run, if everything works as expected, Commissioning Mode can be disabled when post processing.

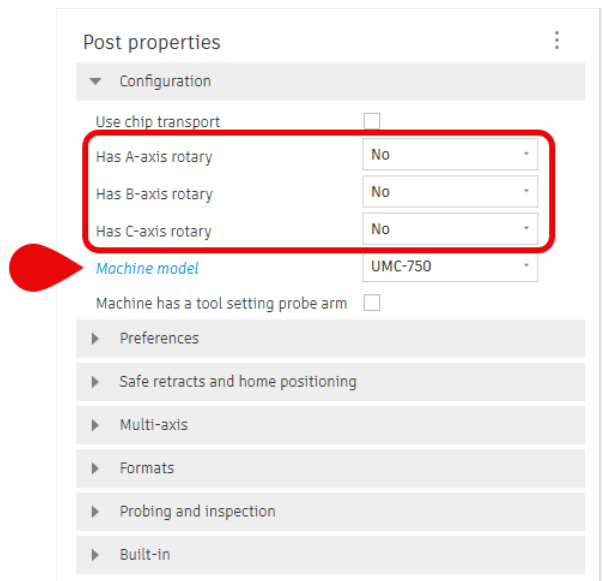
Tutorial:

Follow the below link to see how to use surface inspection with a Haas Next Generation Control.

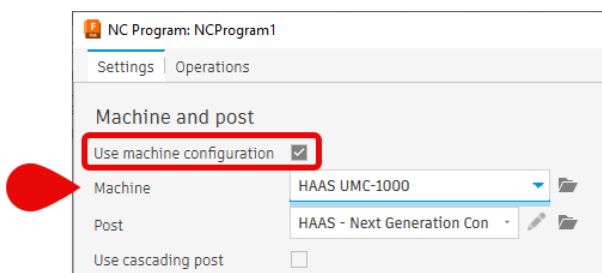
<https://www.youtube.com/watch?v=d9VnOJoQbvg>

Settings for Multi-axis Probing

For 3+2 probing operation the machine kinematics needs to be set on the post processor. While post processing, the Machine model or Rotary axes can be specified in the post property dialog as shown below.



Or else the corresponding machine configuration file can be selected while post processing.



Refer to the following forum link for instructions on how to set the post as multi-axis.

[How to set up a 4/5 axis machine configuration](#)

Reach out to us via [HSM Post Processor Forum - Autodesk Community](#) if you have any issues.